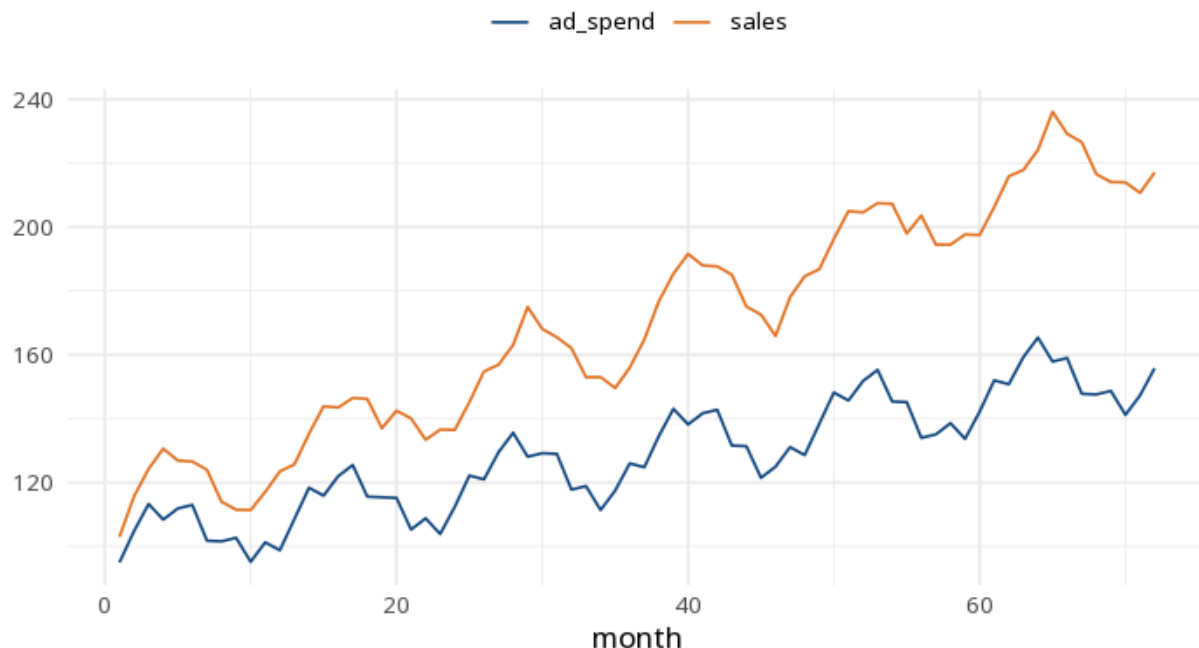


1 Granger causality

Direction	F	df	p
X→Y	47.89	4, 63	<.001
Y→X	30.62	4, 63	<.001



A significant $X \rightarrow Y$ p means past values of X help predict Y beyond Y's own past — this is predictive precedence, not proof that X causes Y. Check both directions. The result depends entirely on the lag order (this card uses a max lag of 4): choose it on theory or by an information criterion — `vars::VARselect` reports the AIC- and BIC-minimising lag — and report it, since different lags can flip the conclusion. Make both series stationary first. Method: `R lmtest::grangertest` (Granger, 1969). Your significance threshold (α) is 0.05.

Granger test $X \rightarrow Y$: $F(4,63)=47.89$, $p < .001$; $Y \rightarrow X$: $F(4,63)=30.62$, $p < .001$ (lag=4).